

**CLASSROOM CONTACT PROGRAMME**
(Academic Session : 2019 - 2020)NEET(UG)
MAJOR
15-02-2020**PRE-MEDICAL : ENTHUSIAST COURSE (ALL PHASE)***This Booklet contains 24 pages.*

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet and fill in the particulars on Side-1 and Side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours duration and this Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
3. Use Blue/Black Ball Point Pen only for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is *not* permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

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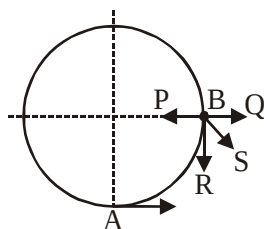
Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2020

1. The displacement x of a particle moving in one dimension under the action of constant force is related to time t by the equation $t = \sqrt{x} + 3$, where x is in metres and t is in seconds. Find the displacement of the particle when its velocity is zero :-

(1) zero (2) 12m (3) 6m (4) 18m

2. A ball of mass m moves inside a smooth spherical shell of radius R with velocity $\sqrt{2gR}$ at 'A'. What is the direction of force acting on the ball. When it reaches B?



(1) BP (2) BQ (3) BR (4) BS

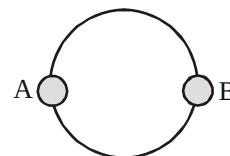
3. In Newton's law of cooling $\frac{d\theta}{dt} = -k\Delta\theta$, the proportionality constant k is k_1 and k_2 for two substances A and B having mass m_1 and m_2 area A_1 and A_2 , specific heat s_1 and s_2 , emissivity e_1 and e_2 respectively, it is given that $\frac{e_1}{e_2} = 1$, $\frac{A_1}{A_2} = 1$,

$\frac{m_1}{m_2} = 2$, $\frac{s_1}{s_2} = \frac{1}{3}$, while surrounding temperature

remains constant. Then $\frac{k_1}{k_2}$:-

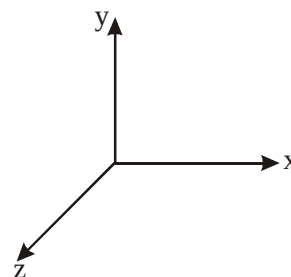
(1) $\frac{1}{3}$ (2) $\frac{2}{9}$
(3) $\frac{3}{2}$ (4) $\frac{3}{4}$

4. A wire of resistance $12 \Omega m^{-1}$ is bent to form a complete circle of radius 10 cm. The resistance between its two diametrically opposite points, A and B as shown in the figure, is



(1) $0.6 \pi \Omega$ (2) 3Ω
(3) $6 \pi \Omega$ (4) 6Ω

5. Light wave is travelling along y-direction. If the corresponding $\vec{E}$ vector at any time is along the x-axis, the direction of $\vec{B}$ vector at that time is along:-



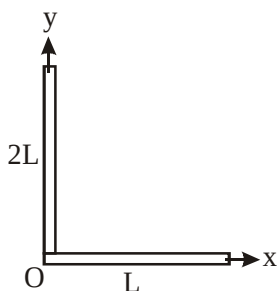
(1) y-axis (2) x-axis (3) +z-axis (4) -z-axis

6. Match the following

(A) Capacitance	(i) Volt (ampere) ⁻¹
(B) Magnetic induction	(ii) Volt-sec(ampere) ⁻¹
(C) Inductance	(iii) Newton (ampere) ⁻¹ (metre) ⁻¹
(D) Resistance	(iv) Coulomb ² (joule) ⁻¹

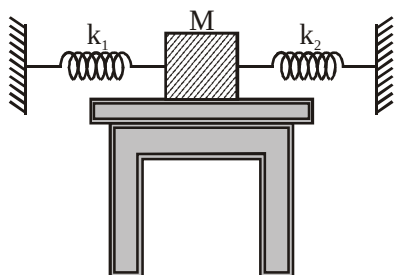
(1) A-ii, B-iii, C-iv, D-i
(2) A-iv, B-iii, C-ii, D-i
(3) A-iii, B-iv, C-i, D-ii
(4) A-iv, B-i, C-ii, D-iii

7. Two uniform rods of length L and $2L$ of same material are placed along x and y as shown in figure than the coordinates of centre of mass of system of rods with respect to point O is :-



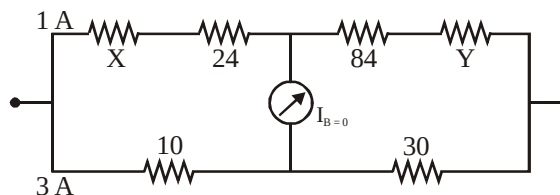
- (1) $\left(\frac{L}{3}, \frac{2L}{3}\right)$ (2) $\left(\frac{L}{4}, \frac{2L}{3}\right)$
 (3) $\left(\frac{L}{6}, \frac{2L}{3}\right)$ (4) $\left(\frac{L}{6}, \frac{L}{3}\right)$

8. A block of mass M is placed on a smooth table. Its two sides are attached to fixed walls by means of collinear horizontal springs of spring constants k_1 and k_2 ($k_1 > k_2$) as shown in the figure. The block is made to oscillate horizontally along the line of the two springs. The frequency of its oscillation is :-



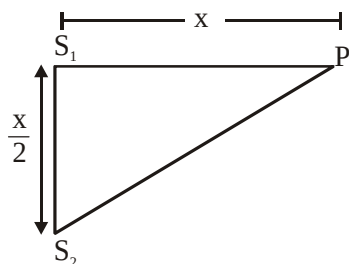
- (1) $\frac{1}{2\pi} \left(\frac{k_1 + k_2}{M} \right)^{1/2}$
 (2) $\frac{1}{2\pi} \left(\frac{k_1 - k_2}{M} \right)^{1/2}$
 (3) $\frac{1}{2\pi} \left(\frac{M}{k_1 + k_2} \right)^{1/2}$
 (4) $\frac{1}{2\pi} \left(\frac{M}{k_1 - k_2} \right)^{1/2}$

9. In the given circuit, the resistances are given in ohm. The current through the 10Ω resistance is 3 A while that through the resistance X is 1 A . No current passes through the galvanometer. The values of the unknown resistances X and Y are respectively (in ohm) :-



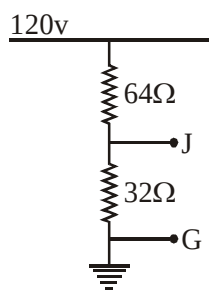
- (1) 14 and 54
 (2) 12 and 6
 (3) 6 and 12
 (4) 6 and 6
10. An object is placed in front of a convex mirror at a distance of 50 cm . A plane mirror is introduced covering lower half of the mirror. If the distance between the object and the plane mirror is 30 cm , it is found that there is no parallax between the images formed by two mirrors. Then radius of curvature of convex mirror :-
- (1) 30 cm (2) 40 cm
 (3) 25 cm (4) 15 cm
11. Data for four screwgauge's given below, arrange them increasing order of accuracy.
- (a) Pitch = 1 mm , no. of division on C.S. = 100
 (b) Pitch = 0.5 mm , no. of division on C.S. = 100
 (c) Pitch = $3/4 \text{ mm}$, no. of division on C.S. = 60
 (d) Pitch = $1/2 \text{ mm}$, no. of division on C.S. = 50
- (1) $b < d = a < c$ (2) $a = d < b < c$
 (3) $c < d = a < b$ (4) $a = d < c < b$
12. A neutron moving with velocity u collides elastically with an nucleus of mass number A . If the collision is head on and the initial kinetic energy of neutron is E , then the final kinetic energy of the neutron after collision is :
- (1) $\left(\frac{A+1}{A-1}\right)^2 E$ (2) $\left(\frac{A-1}{A+1}\right)^2 E$
 (3) $\left(\frac{A-1}{A+1}\right) E$ (4) $\left(\frac{A+1}{A-1}\right) E$

13. The wavelength of the waves arriving at P from two coherent sources S_1 and S_2 in 4m while intensity of each wave is I_0 . The resultant intensity at P is $2I_0$. Find minimum value of S_2P :



- (1) $\frac{x\sqrt{5}}{2}$ (2) $2x$
(3) $x + 1$ (4) $(x + \sqrt{2})$

14. Find the potential of J with respect of 'G' :-



- (1) 40 V (2) 60 V (3) 20 V (4) 30 V

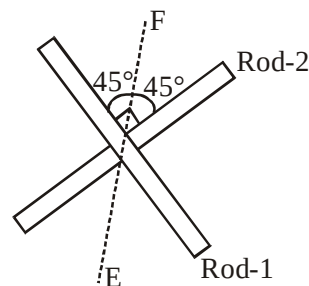
15. A person can see upto distance x, wants to see distance y ($y > x$) then power of the lens should be:-

- (1) $\frac{xy}{x-y}$ (2) $\frac{x}{y}$ (3) $\frac{y}{x}$ (4) $\frac{x-y}{xy}$

16. The motion of a particle along a straight line is described by the equation: $x = 8 + 12t - t^3$, where x is in meter and t in second.

- (i) the initial velocity of particle is 12 m/s
(ii) the retardation of particle when velocity is zero is 12 m/s^2
(iii) when acceleration is zero, displacement is 8 m
(iv) the maximum velocity of particle is 12 m/s
(1) (i), (ii)
(2) (ii), (iii)
(3) (i), (ii), (iii)
(4) All options are correct

17. Two uniform, thin identical rods each of mass M and length L are joined at middle so as to form a cross as shown. The moment of inertia of the cross about a bisector line EF (in the plane of cross) is:

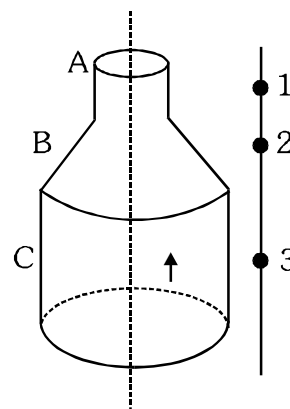


- (1) $\frac{ML^2}{6}$ (2) $\frac{ML^2}{4}$ (3) $\frac{ML^2}{12}$ (4) $\frac{ML^2}{3}$

18. A bus is moving with a velocity of 5 m/s towards a huge wall. The driver sounds a horn of frequency 165 Hz. If the speed of sound in air is 335 m/s, the number of beats heard per second by the passengers in the bus will be :-

- (1) 3 (2) 4 (3) 5 (4) 6

19. A long, straight, hollow conductor tube carrying a current has two sections A and C of unequal cross section joined by a conical section B. 1, 2 and 3 are the points on a line parallel to the axis of the conductor. The magnetic fields at 1, 2 and 3 have magnitude B_1 , B_2 and B_3 :-

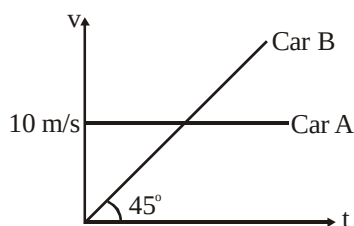


- (1) $B_1 = B_2 = B_3$ (2) $B_1 = B_2 \neq B_3$
(3) $B_1 < B_2 < B_3$ (4) $B_1 > B_2 > B_3$

20. The path difference between two interfering waves of equal intensities at a point on the screen is $\lambda/4$. The ratio of intensity at this point and that at the central fringe will be

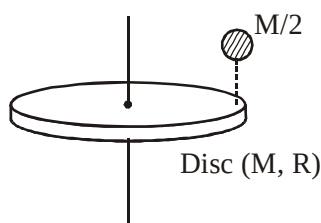
- (1) 1 : 1 (2) 1 : 2
(3) 2 : 1 (4) 1 : 4

21. Initially car A is 10.5 m ahead of car B. Both start moving at time $t = 0$ in the same direction along a straight line. The velocity-time graph of two cars is shown in figure. The time when the car B will catch the car A, will be :-



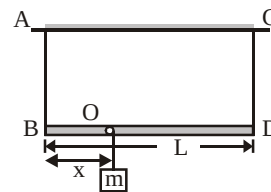
- (1) $t = 21$ sec (2) $t = 2\sqrt{5}$ sec
(3) 20 sec (4) none

22. A horizontal disc (M, R) is rotating about its own axis with angular velocity ω_0 . Another particle of mass $M/2$ is gently placed at its circumference which gets stuck to it. Find the final angular velocity of disc.



- (1) $\frac{\omega_0}{3}$ (2) $\frac{2\omega_0}{3}$
(3) $\frac{\omega_0}{2}$ (4) $\frac{\omega_0}{6}$

23. A massless rod is suspended by two identical strings AB and CD of equal length. A block of mass m is suspended from point O such that BO is equal to 'x'. Further, it is observed that the frequency of 1st harmonic (fundamental frequency) in AB is equal to 2nd harmonic frequency in CD. Then, length of BO is :-

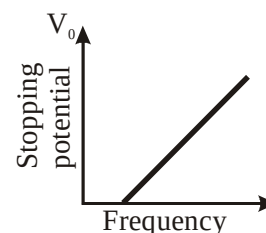


- (1) $\frac{L}{5}$ (2) $\frac{4L}{5}$
(3) $\frac{3L}{4}$ (4) $\frac{L}{4}$

24. A test charge 1.6×10^{-19} C is moving with velocity $\vec{v} = (2\hat{i} + 3\hat{j})$ m/sec in a magnetic field $\vec{B} = (4\hat{i} + 6\hat{j})$ Wb/m². The magnetic force on the test charge :-

- (1) $6\hat{k}\text{T}$ (2) $(8\hat{i} + 18\hat{j})\text{T}$
(3) $(4\hat{i} + 6\hat{j}) \times 10^{-19}\text{T}$ (4) zero

25. In the graph given below. If the slope is 4.12×10^{-15} V-s, then value of 'h' should be :-

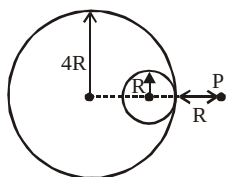


- (1) 6.6×10^{-31} J-s
(2) 6.6×10^{-34} J-s
(3) 9.1×10^{-31} J-s
(4) 6.6×10^{-32} J-s

26. An empty plastic box of mass m is found to accelerate up at the rate of $g/6$ when placed deep inside water. How much mass of sand should be put inside the box so that it may accelerate down at the rate of $g/6$:-

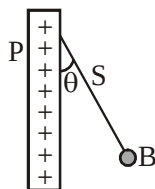
- (1) $\frac{2m}{3}$ (2) $\frac{2m}{5}$
(3) $\frac{3m}{4}$ (4) $\frac{3m}{5}$

27. A cavity of radius R is made in the solid sphere of uniform density ρ & radius $4R$. Find gravitational intensity at point P :-



- (1) $\frac{231}{75}G\rho\pi R$ (2) $\frac{256}{75}G\rho\pi R$
(3) $\frac{256}{25}G\rho\pi R$ (4) $\frac{1}{3}G\rho\pi R$

28. A charged ball B hangs from a silk thread S , which makes an angle θ with a large charged conducting sheet P , as shown in the figure. The surface charge density σ of the sheet is proportional to :-



- (1) $\sin \theta$ (2) $\tan \theta$
(3) $\cos \theta$ (4) $\cot \theta$

29. At a place horizontal and vertical components of earth's magnetic field are as follows

$$B_H = \sqrt{3} \text{ G } 10^\circ \text{ west of north}$$

$$B_V = 1 \text{ G vertically downward}$$

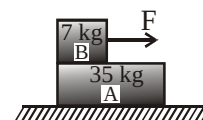
Then dip angle and declination are respectively :-

- (1) $0^\circ, 10^\circ \text{ W}$
(2) $30^\circ, 10^\circ \text{ E}$
(3) $45^\circ, 10^\circ \text{ W}$
(4) $30^\circ, 10^\circ \text{ W of N}$

30. An electron with speed v and a photon with speed c have the same de Broglie wavelength. If the kinetic energy and momentum of electrons is E_e and P_e and that of photon is E_{ph} and P_{ph} respectively, the correct option is :-

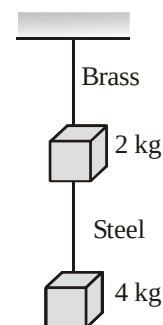
- (1) $\frac{E_e}{E_{ph}} = \frac{2c}{v}$
(2) $\frac{E_e}{E_{ph}} = \frac{v}{2c}$
(3) $\frac{P_e}{P_{ph}} = \frac{2c}{v}$
(4) $\frac{P_e}{P_{ph}} = \frac{v}{2c}$

31. Block A of mass 35 kg is resting on a frictionless floor. Another block B of mass 7 kg is resting on it as shown in the figure. The coefficient of friction between the blocks is 0.5 while kinetic friction is 0.4. If a force of 100 N is applied to block B, the acceleration of the block A will be : ($g = 10 \text{ ms}^{-2}$) :-



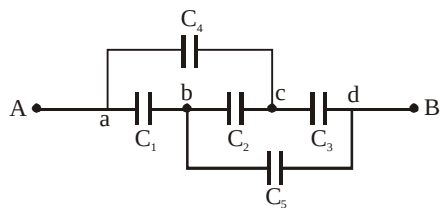
- (1) 0.8 ms^{-2} (2) 2.4 ms^{-2}
(3) 0.4 m s^{-2} (4) 4.4 m s^{-2}

32. If the ratio of length, radii and Young's modulus of steel and brass wires shown in the figure are a , b and c , respectively. The ratio between the increase in lengths of brass and steel wires would be :-

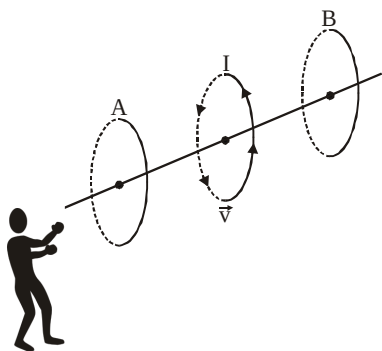


- (1) $\frac{b^2a}{2c}$ (2) $\frac{bc}{2a^2}$
(3) $\frac{3b^2c}{2a}$ (4) $\frac{a}{2b^2c}$

33. In the given figure, the capacitors C_1 , C_3 , C_4 and C_5 have a capacitance of $4\mu\text{F}$ each. If the capacitor C_2 has a capacitance of $10\mu\text{F}$, then the effective capacitance between A and B is :-

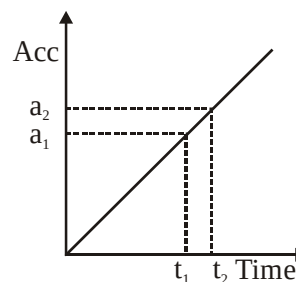


- (1) $2\mu\text{F}$ (2) $4\mu\text{F}$ (3) $6\mu\text{F}$ (4) $8\mu\text{F}$
34. Three coaxial circular wire loops and an stationary observer are positioned as shown in figure. From the observers point of view, a current I flows counter clockwise in the middle loop, which is moving towards the observer with a velocity v . Loops A and B are stationary. The same observer would notice that :

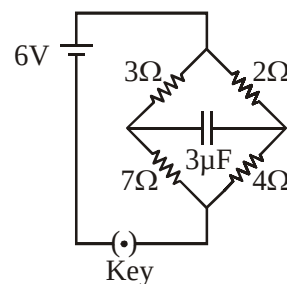


- (1) Clockwise currents are induced in loops A and B.
 (2) Counter clockwise currents are induced in loops A and B.
 (3) A clockwise current is induced in loop A, but a counter clockwise current is induced in loop B.
 (4) A counter clockwise current is induced in loop A, but a clockwise current is induced in loop B.
35. The half-life of a radioactive substance is 30 minutes. The time (in minutes) taken between 40% decay and 85% decay of the same radioactive substance is :
- (1) 45 (2) 60
 (3) 15 (4) 30

36. Acceleration time graph of a particle is shown. Work done by all the forces acting on the particle of mass m in time interval t_1 and t_2 while a_1 is the acceleration at time t_1 , is given by :
 (at $t = 0$, particle was at rest):-

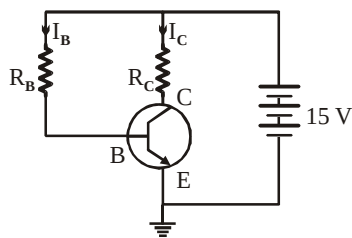


- (1) $\frac{ma_1^2}{4t_1}(t_2^3 - t_1^3)$ (2) $\frac{ma_1^2}{8t_1^2}(t_2^4 - t_1^4)$
 (3) $\frac{ma_1^2}{4t_1^2}(t_2^4 - t_1^4)$ (4) $\frac{ma_1}{2t_1}(t_2^2 - t_1^2)$
37. An air bubble of radius r in water is at a depth h below the water surface at some instant. If P is atmospheric pressure, d and T are density and surface tension of water respectively, the pressure inside the bubble will be :
- (1) $P + h dg - (4T/r)$ (2) $P + h dg + (2T/r)$
 (3) $P + h dg - (2T/r)$ (4) $P + h dg + (4T/r)$
38. In steady state; energy stored in the capacitor is:-

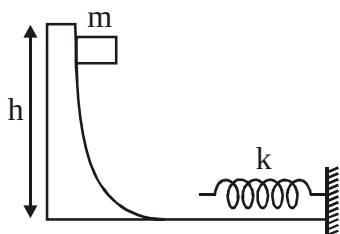


- (1) $80 \times 10^{-9} \text{ J}$ (2) $20 \times 10^{-9} \text{ J}$
 (3) $100 \times 10^{-9} \text{ J}$ (4) $60 \times 10^{-9} \text{ J}$

39. The average emf induced in a coil when a current changes from 0 to 2 A in 0.05 sec is 8V. The self-inductance of the coils is :-
- (1) 0.1 H (2) 0.2 H
(3) 0.4 H (4) 0.8 H
40. In the following common emitter circuit if $\beta = 100$, $V_{CE} = 7V$, $V_{BE} = \text{Negligible}$, $R_C = 2k\Omega$, then $I_B = ?$

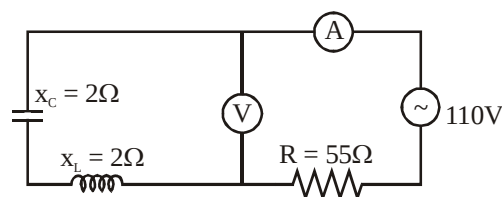


- (1) 0.01 mA
(2) 0.04 mA
(3) 0.02 mA
(4) 0.03 mA
41. A block of mass m is released from a height h from the top of a smooth surface. There is an ideal spring of spring constant k at the bottom of the track. Find the maximum compression in the spring (Wedge is fixed) :-

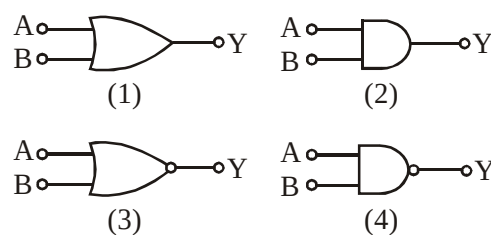


- (1) $\sqrt{\frac{2mgh}{k}}$ (2) $\sqrt{\frac{mgh}{k}}$
(3) $\sqrt{\frac{mgh}{2k}}$ (4) Cannot be determined

42. A perfect gas from state A to another state B by absorbing 8×10^5 J of heat and receiving 6.5×10^5 J of external work. If it is now transferred between the same two states in another process in which it absorbs 10^5 J of heat. Then in the second process :-
- (1) Work done on the gas is 0.5×10^5 J
(2) Work done by gas is 0.5×10^5 J
(3) Work done on gas is 10^5 J
(4) Work done by gas is 10^5 J
43. Two uniform wires A and B are of the same metal and have equal masses. The radius of wire A is twice that of wire B. The total resistance of A and B when connected in parallel is :-
- (1) 4Ω when the resistance of wire A is 4.25Ω
(2) 5Ω when the resistance of wire A is 4.25Ω
(3) 4Ω when the resistance of wire B is 4.25Ω
(4) 5Ω when the resistance of wire B is 4.25Ω
44. In the given figure as shown the reading of the ammeter in ampere is:



- (1) 2 (2) 3
(3) Zero (4) 1
45. Given below are four logic gate symbols. Those for OR, NOR and NAND are respectively :-



- (1) 1, 4, 3 (2) 4, 1, 2
(3) 1, 3, 4 (4) 4, 2, 1

46. In which of the following reaction H_2SO_4 acts as oxidising agent :-

- (1) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{CaSO}_4 + 2\text{HF}$
 (2) $2\text{NaCl} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl}$
 (3) $\text{Cu} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$
 (4) $\text{C}_{12}\text{H}_{22}\text{O}_{11} \xrightarrow{\text{Conc. H}_2\text{SO}_4} 12\text{C} + 11\text{H}_2\text{O}$

47. Interhalogen compound which is found in dimeric form is :-

- (1) BrF_6 (2) IF_7 (3) ICl (4) ICl_3

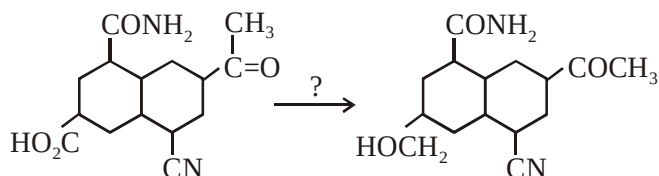
48. Heat exchanged during the boiling of water is $0.900 \text{ KJ/mol}^{-1}$. What will be the increase in entropy.

- (1) $2.41 \text{ JK}^{-1} \text{ mol}^{-1}$ (2) $87.2 \text{ JK}^{-1} \text{ mol}^{-1}$
 (3) $900 \text{ JK}^{-1} \text{ mol}^{-1}$ (4) Zero

49. The equimolar mixture of CH_3COONa and CH_3COOH shows a pH equal to 5. Then the K_a of the CH_3COOH will be

- (1) 10^{-10} (2) 10^{-5}
 (3) 5×10^{-5} (4) 5×10^{-10}

50. The most suitable reagent for the given conversion is:



- (1) (i) LiAlH_4 (ii) H_2O (2) (i) NaBH_4 (ii) H_2O
 (3) H_2/Pd (4) (i) B_2H_6 (ii) H_2O

51. Which pair of elements stabilize the higher oxidation states ?

- (1) F, Cl (2) F, O (3) O, Cl (4) O, H

52. $\text{Br}_2 + 2\text{X}_{(\text{aq})}^- \longrightarrow \text{X}_2 + 2\text{Br}_{(\text{aq})}^-$

Above reaction is correct when 'X' is ?

- (1) F^- , Cl^-
 (2) I^- Only
 (3) F^- , Cl^- , I^-
 (4) Not possible for any halide

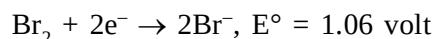
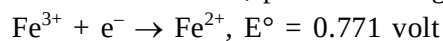
53. Enthalpy of formation of $\text{CaCO}_3(\text{s})$, $\text{CO}_2(\text{g})$, $\text{CaO}(\text{s})$ is -288 KJ , -94.0 KJ and -152 KJ respectively.

Find the enthalpy change of the reaction

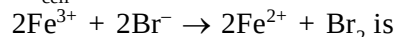


- (1) 42 KJ (2) 21 KJ (3) -42 KJ (4) -21 KJ

54. Standard electrode, potential are given.



E°_{cell} for the cell reaction is :



- (1) $2 \times 0.771 - 1.06 = 0.482 \text{ Volt}$
 (2) $0.771 - 0.5 \times 1.06 = 241 \text{ Volt}$
 (3) $1.06 - 0.771 = 0.289 \text{ Volt}$
 (4) $0.771 - 1.06 = -0.289 \text{ Volt}$

55. The most suitable reagent for the conversion of $\text{RCH}_2\text{-OH} \longrightarrow \text{RCHO}$ is :-

- (1) $\text{KMnO}_4 / \text{H}^+$; Δ
 (2) $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}^+$; Δ
 (3) $\text{CrO}_3 / \text{H}^+$
 (4) PCC (Pyridinium Chlorochromate)

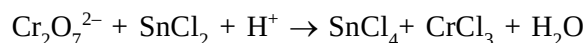
56. On reaction with O_2 , N_2 & H_2O the products formed by lanthanoids are :-

- (1) Ln_2O_3 , Ln_2N_3 & $\text{Ln}(\text{OH})_2$
 (2) Ln_2O_3 , LnN & $\text{Ln}(\text{OH})_3$
 (3) Ln_2O_3 , Ln_2N_3 & $\text{Ln}(\text{OH})_3$
 (4) Ln_2O_3 , LnN & $\text{Ln}(\text{OH})_2$

57. The total number of paramagnetic complex which are inner orbital complexes in given complex:-

- $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$, $[\text{Ni}(\text{NH}_3)_6]\text{SO}_4$, $\text{K}_2[\text{PtCl}_6]$, $[\text{V}(\text{H}_2\text{O})_6]\text{SO}_4$, $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$
 (1) 3 (2) 4 (3) 5 (4) 2

58. In the reaction



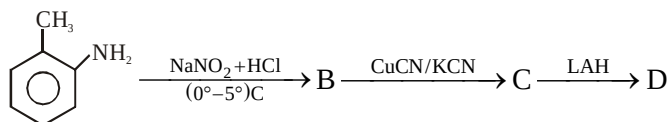
- (1) $\text{Cr}_2\text{O}_7^{2-}$ and H^+ both are reduced
 (2) $\text{Cr}_2\text{O}_7^{2-}$ is reduced and H^+ is oxidized
 (3) $\text{Cr}_2\text{O}_7^{2-}$ is reduced and SnCl_2 is oxidised
 (4) $\text{Cr}_2\text{O}_7^{2-}$ is oxidized and SnCl_2 is reduced

59. 50ml solution of $\frac{\text{N}}{10} \text{CaCl}_2$ is to be diluted to form

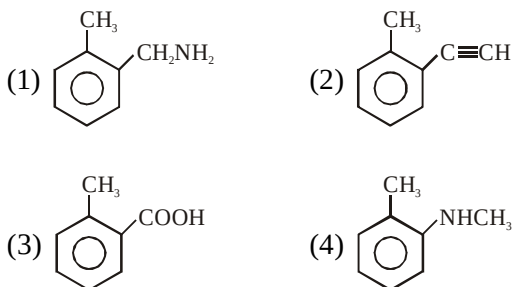
a solution of $\frac{\text{M}}{40}$. What amount of water should be added?

- (1) 100 ml (2) 75ml
 (3) 25ml (4) 50ml

60. In the reaction :



D would be :-



61. Match column I to column II :-

Column (I)		Column (II)	
(a)	Cerrusite	(P)	FeCO_3
(b)	Siderite	(Q)	Ag_2S
(c)	Silver glance	(R)	$\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$
(d)	Limonite	(S)	PbCO_3

- (1) a – P, b – Q, c – R, d – S
 (2) a – S, b – P, c – Q, d – R
 (3) a – S, b – R, c – Q, d – P
 (4) a – R, b – S, c – Q, d – P

62. Which d-orbital of metal in $\text{Fe}(\text{CO})_5$ is involved in hybridisation :-

- (1) dz^2 (2) dxy
 (3) dyz (4) dxz

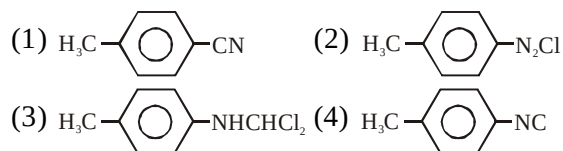
63. Three rubber tube are respectively filled with N_2 , H_2 , CO at same temperature and pressure. The tube which will be reinflated first is

- (1) H_2 filled tube
 (2) CO will filled tube
 (3) N_2 filled tube
 (4) All tube are filled simultaneously.

64. The number of stereoisomers of compound $\text{CH}_3\text{--CH=CH--CH}(\text{Br})\text{CH}_3$ is :-

- (1) 2 (2) 3 (3) 4 (4) 6

65. The reaction of chloroform with alcoholic KOH and p-toluidine form :-



66. SbF_5 reacts with XeF_4 and XeF_6 to form ionic compounds $[\text{XeF}_3^+][\text{SbF}_6^-]$ and $[\text{XeF}_5^+][\text{SbF}_6^-]$. The geometry of XeF_3^+ ion and XeF_5^+ ion, respectively, is :-

- (1) Square pyramidal, T-shaped
 (2) Bent T-shaped, square pyramidal
 (3) See-saw, square pyramidal
 (4) Square pyramidal, see-saw

67. Which of the following is liquid non-metal element?
 (1) Br (2) Ga (3) Hg (4) All

68. A compound consists of atoms A and B crystalizes in a cubic closed packed structure. Atoms of A are located at all the corners and atoms

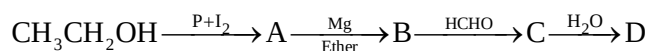
of B are at $\frac{1}{4}$ th of octahedral voids. The simplest formula of the compound will be

- (1) AB (2) A_4B (3) AB_4 (4) A_2B_3

69. Reaction of trans-1-Bromo-2-phenylcyclopentane on reaction with alcoholic KOH produces.

- (1) 1-phenylcyclopentene
 (2) 3-phenylcyclopentene
 (3) 4-phenylcyclopentene
 (4) 2-phenylcyclopentene

70. In the following sequence of reactions,



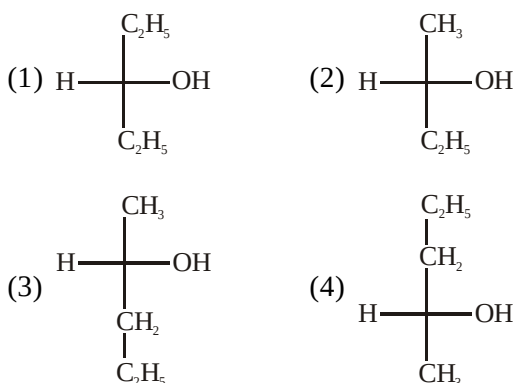
the compound D is :-

- (1) Propanal (2) Butanal
 (3) n-butylalcohol (4) n-propylalcohol

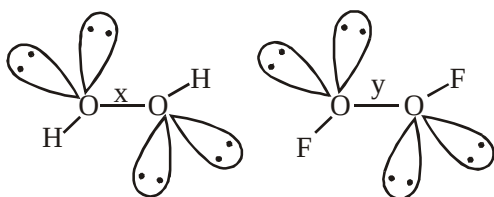
71. The nodal plane in the π -bond of ethene is located in:-

- (1) The molecular plane
 (2) A plane perpendicular to the molecular plane
 (3) A plane perpendicular to the molecular plane which bisects the carbon-carbon sigma bond at right angle
 (4) A plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond

72. In which of the following compounds chromium (Cr) shows minimum radius ?
 (1) CrO (2) Cr₂O₃ (3) CrO₃ (4) None
73. Which is not a condensation method for preparing the colloidal sol ?
 (1) Double decomposition
 (2) Hydrolysis
 (3) Peptization
 (4) Oxidation
74. Which of the following is the structure of (S)-Pentan-2-ol is ?



75. A compound C₅H₈, which give white ppt. with ammonical AgNO₃. A give (CH₃)₂CHCOOH with hot alkaline KMnO₄ then compound is :-
 (1) CH₃CH₂-CH₂-CH=CH₂
 (2) CH₃-CH₂-C≡CH
 (3) (CH₃)₂CH-C≡CH
 (4) CH₂=CH-CH₂-CH=CH₂
76. Compare bond lengths for the following molecules (O-O B.L.) :-

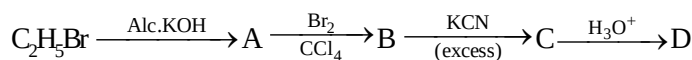


- (1) x > y (2) y > x
 (3) x = y (4) None of these
77. Relative density of a volatile substance with respect to O₂ is 8. Its molecular weight would be
 (1) 64 (2) 128 (3) 256 (4) 80

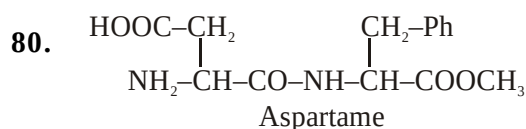
78. Given that $E_{Ag^+/Ag}^0 = 0.80V$
 $E_{Mg^{2+}/Mg}^0 = -2.30V$
 $E_{Cu^{2+}/Cu}^0 = 0.34V$
 $E_{Cr^{3+}/Cr}^0 = -0.75V$

Which of the following statement is true :

- (1) AgNO₃ solution can be stored in copper container
 (2) Mg(NO₃)₂ solution can not be stored in copper container.
 (3) CuCl₂ solution can be stored in silver container
 (4) CrCl₃ solution can be stored in magnesium container.
79. The acid D obtained through the following sequence of reactions is :-



- (1) Succinic acid (2) Malonic acid
 (3) Maleic acid (4) Oxalic acid



Aspartame is 100 times as sweet as sucrose and is used as a sugar substitute. The correct statement(s) about aspartame is (are)

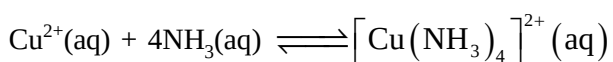
- (I) It is an ester derivative of dipeptide
 (II) It can be named as aspartyl phenyl alanine methyl ester
 (III) It is a tripeptide
 (IV) It is having four functional groups
- (1) I, II (2) I, II, IV
 (3) II, III, IV (4) Only II
81. Which of the following correct ?
 (1) KMnO₄ is violet / purple / dark pink due to d-d transition.
 (2) TiF₆⁻³ & Cu₂Cl₂ both are colorless
 (3) AgCl is coloured due to polarisation
 (4) [Co(NH₃)₆]²⁺ < [Co(H₂O)₆]²⁺ λ of absorbed light
82. Maximum number of electron in L shell having spin quantum number $s = -\frac{1}{2}$ is
 (1) 4 (2) 8 (3) 2 (4) 6

83. 0.25g of non electrolyte compound is dissolved in 25g of a solvent. Which shows an elevation in boiling point of 0.216°C . The molar mass of the compound will be [K_b of solvent = 2.16°C]
 (1) 50 (2) 100 (3) 75 (4) 125
84. Match the chemical conversion in column-I with the appropriate reagents in column-II and select the correct answer using the code given below the lists:-

Column-I		Column-II	
(A)		X	(i) $\text{Hg}(\text{OAc})_2 + \text{H}_2\text{O}$; (ii) NaBH_4
(B)		Y	$\text{NaOEt} + \Delta$
(C)		Z	Et-Br
(D)		W	(i) $\text{BH}_3 + \text{T.H.F.}$ (ii) $\text{H}_2\text{O}_2/\text{NaOH}$

- | A | B | C | D |
|-------|---|---|---|
| (1) Y | Z | X | W |
| (2) Z | Y | X | W |
| (3) Y | Z | W | X |
| (4) Z | Y | W | X |

85. Which of the following is a biodegradable polymer?
 (1) PHBV (2) Nylon-6
 (3) Bunna-S (4) Buna-N
86. Select correct about covalent character :-
 (1) $\text{SnCl}_4 > \text{SnCl}_2$ (2) $\text{SbCl}_5 > \text{SbCl}_3$
 (3) $\text{UF}_6 > \text{UF}_4$ (4) all of the above
87. For the reaction

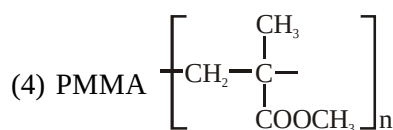
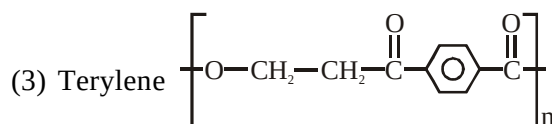
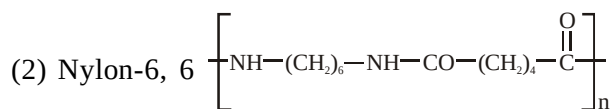
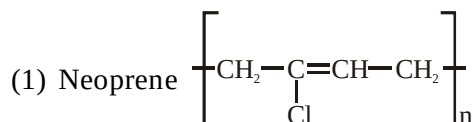


Value of K_C is 10^{10} at temperature 25°C . The value of its K_p will be :

- (1) $10^{10} \times (22.4)^2$
 (2) $\frac{10^{10}}{(22.4)^2}$
 (3) 10^{10}
 (4) Value of K_p can not be found.

88. In a reaction $\text{A} \rightarrow \text{B}$, A decomposes to 50% in 7min, 75% in 14min. and 87.5% in 21 min. The reaction follows the
 (1) Zero order kinetics
 (2) First order kinetics
 (3) Second order kinetics
 (4) $\frac{3}{2}$ th order kinetics

89. Which one of the following is not correctly matched?

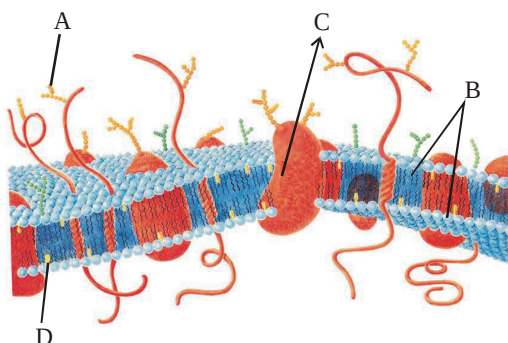


90. Which of the following statement about detergent is incorrect ?
 (1) Detergents containing branched hydrocarbon chains are biodegradable
 (2) Cationic detergents are quaternary ammonium salts of amines with acetates, chlorides or bromides as an ions
 (3) Anionic detergents are sodium salts of sulphonated long chain alcohols or hydrocarbons
 (4) Non-ionic detergents do not contain any ion in their constitution

91. In which one of the following, the genus name, its character and its class are not correctly matched, whereas the remaining three are correctly matched?

	Genus Name	Character	Class
(1)	<i>Alternaria</i>	Sexual stage is absent	Deuteromycetes
(2)	<i>Neurospora</i>	Endogenous ascospores	Ascomycetes
(3)	<i>Pythium</i>	Dikaryotic mycelium	Basidiomycetes
(4)	<i>Albugo</i>	Aseptate and coenocytic mycelium	Phycomycetes

92. A simple mechanical tissue with high refractive tissue, present below epidermis either as a homogenous layer or in patches, whose cell walls of cells show pectocellulosic unignified thickenings often at corners, which is primary in nature and provide tensile strength to various growing plant organs
- (1) Parenchyma (2) Collenchyma
(3) Sclerenchyma (4) Secondary xylem
93. Study the following diagram carefully and select the correct statement regarding the diagram.



- (1) Structure A is integral protein & can not be removed easily
(2) Structure C is integral protein and can be removed easily
(3) Structure D is phospholipid bilayer which is made up of hydrophilic molecule
(4) Structure C is integral protein and cannot be removed easily

94. Which of the following is correct for C_3 & C_4 plants regarding number of carbons (C) in the primary CO_2 acceptor & number of carbons (C) in the primary CO_2 fixation product, respectively?

- (1) C_3 plant $\Rightarrow$ 3C, 3C
 C_4 plant $\Rightarrow$ 4C, 4C
(2) C_3 plant $\Rightarrow$ 5C, 3C
 C_4 plant $\Rightarrow$ 3C, 4C
(3) C_3 plant $\Rightarrow$ 5C, 3C
 C_4 plant $\Rightarrow$ 4C, 4C
(4) C_3 plant $\Rightarrow$ 3C, 3C
 C_4 plant $\Rightarrow$ 3C, 4C

95. A frog with body temperature of $20^\circ C$ is transferred to an area with $30^\circ C$ temperature. What will be the body temperature of frog in the new environment?

- (1) $20^\circ C$
(2) $30^\circ C$
(3) $25^\circ C$
(4) Fluctuates between $20-30^\circ C$

96. Main function of cerebellum is

- (1) Balancing (2) To see
(3) To hear (4) Remembering

97. In an aquatic ecosystem _____ is the main conduit of energy flow.

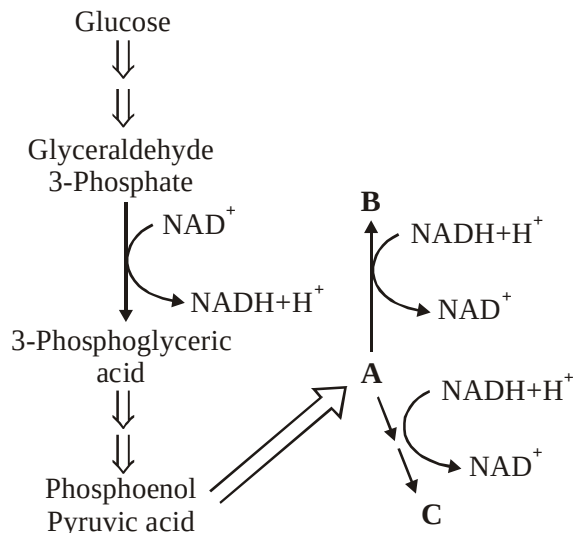
- (1) GFC
(2) DFC
(3) Both (1) & (2)
(4) Parasitic food chain

98. Occurrence of more than one embryo in a seed is called.

- (1) Parthenogenesis (2) Parthenocarpy
(3) Polyembryony (4) Apogamy

99. Which factor not affect the hardy-weinberg equilibrium?
- Gene migration and mutation
 - Genetic recombination and natural selection
 - Genetic drift, gene flow
 - large population and random mating
100. Which of the following hormone is secreted by ovary as well as placenta-
- Progesterone
 - Estrogen
 - Relaxin
 - All of above
101. Archaeobacteria can survive in adverse habitat due to
- The presence of special type protein in cell membrane
 - The presence of special type protein in protoplasm
 - Unique structure of cell wall and cell membrane
 - Homopolar bonds in proteins
102. Select incorrect statement ?
- Bast fibre of jute, flax and hemp are used commercially.
 - In leaves, the ground tissue consists of thin-walled chloroplast containing cells and is called mesophyll.
 - In a woody dicotyledonous stem, oldest layer of secondary xylem is located just above primary xylem.
 - In the internal structure of dorsiventral leaf, position of xylem in the vascular bundle is abaxial
103. Read the following statement carefully and identify the two incorrect statement.
- 70s ribosome are present in prokaryotic and eukaryotic cell both
 - Flagella are long structure causing the movement of either the cell or surrounding fluid
 - Microtubule of cilia, flagella and centriole are made up of actin protein
 - Chromatin material in eukaryotes are consist of DNA, RNA, histone and non histon protein
- Option :-
- A & D
 - B & C
 - A & C
 - B & D

104.



Above figure showing the major pathways of anaerobic respiration/fermentation. Which of the following is correct regarding enzymes involved in conversion of A into B and A into C?

- A → B ⇒ Pyruvate dehydrogenase & alcohol dehydrogenase
A → C ⇒ Lactate dehydrogenase
 - A → B ⇒ Lactate dehydrogenase
A → C ⇒ Pyruvate dehydrogenase
 - A → B ⇒ Lactate dehydrogenase
A → C ⇒ Pyruvic acid decarboxylase and alcohol dehydrogenase
 - A → B ⇒ Pyruvic acid decarboxylase
A → C ⇒ Alcohol dehydrogenase
105. Which of the following features is used to identify a male cockroach from a female cockroach.
- Presence of a boat shaped sternum on the 9th abdominal segment.
 - Presence of caudal styles
 - Forewings with darker tegmina
 - Presence of anal cerci
106. Contraction of involuntary muscles, secretion of digestive glands and rate of heart beats are under the control of
- Cranial system
 - Reflex system
 - Autonomic nervous system
 - Central nervous system

107. Select the incorrect pair :-

- (1) Gaseous nutrient cycle – Carbon cycle
- (2) Sedimentary nutrient cycle – Phosphorus cycle
- (3) Gaseous nutrient cycle – Nitrogen cycle
- (4) Sedimentary nutrient cycle – Carbon cycle

108. Which of the following pair is correct?

- (1) Endospermic seed – Pea, Maize
- (2) Non endospermic seed – Pea, Groundnut
- (3) Parthenocarpic fruit – Coconut, Banana
- (4) False fruit – Tomato, Apple

109. Evolution below species level is known as

- (1) Micro evolution (2) Macro evolution
- (3) Mega evolution (4) Natural selection

110. In which technique embryo upto 8 cell stage are transferred into fallopian tube ?

- (1) I.U.T. (2) G.I.F.T.
- (3) Z.I.F.T. (4) A.I.

111.	S.N.	Character	Bryophyta	Pteridophyta	Gymnosperms
	(I)	Main plant body	Gametophyte	Sporophyte	(A)
	(II)	Vascular tissue	Absent	(B)	Present
	(III)	Embryo formation	(C)	Occurs	Occurs
	(IV)	(D)	Archegonium	Archegonium	Archegonium

Identify A, B, C & D in the above table :-

	A	B	C	D
(1)	Sporophyte	Absent	Does not occur	Female reproductive organ
(2)	Gametophyte	Absent	Does not occur	Female reproductive organ
(3)	Sporophyte	Present	Occurs	Female reproductive organ
(4)	Sporophyte	Present	Does not occur	Male reproductive organ

112. Which of the following is a part of periderm?

- (1) Cork (2) Secondary xylem
- (3) Vascular cambium (4) Spring wood

113. Synthesis of protein, lipoproteins and glycogen occurs in:

- (1) Golgibody (2) Endoplasmic reticulum
- (3) Lysosome (4) Chloroplast

114. In plants :

- (1) growth is open & differentiation is closed
- (2) differentiation is open & growth is closed
- (3) both growth & differentiation are closed
- (4) both growth & differentiation are open

115. What is common among Scoliodon, Whale, Bat and Rat?

- (1) Presence of muscular diaphragm between thorax and abdomen
- (2) Presence of middle ear
- (3) Presence of arteriovenous heart
- (4) Absence of external fertilization and indirect development

116. Which of the following cranial nerves are involved in the movement of eye ball ?

- (1) Optic, oculomotor, abducens
- (2) Oculomotor, abducens, trochlear
- (3) Trochlear, abducens and optic
- (4) Abducens, optic, trochlear, oculomotor

117. A gene locus has two alleles A, a. If the frequency of dominant allele A is 0.2 then what will be the frequency of homozygous dominant, heterozygous and homozygous recessive individuals in the population :

- (1) 0.36(AA), 0.64(Aa) ; 0.16(aa)
- (2) 0.40(AA), 0.64(Aa) ; 0.32(aa)
- (3) 0.04(AA), 0.32(Aa) ; 0.64(aa)
- (4) 0.32(AA), 0.04(Aa) ; 0.64(aa)

118. Which hormone is administered in cattles to induce follicular maturation and superovulation

- (1) Prolactin (2) FSH
- (3) Oxytocin (4) ADH

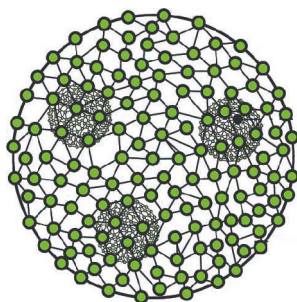
119. Find out the correct sequence of evolution of birds

- (1) Early reptiles–sauropsids–therapsid–bird
- (2) Early reptiles–synapsids–thecodont–birds
- (3) Early reptiles–sauropsids–thecodont–birds
- (4) Early reptiles–synapsids–sauropsids–birds

120. The sexually transmitted infection [STI] caused by a protozoan is :-

- (1) Gonorrhoea (2) Syphilis
- (3) Trichomoniasis (4) Chlamydia

121. Study the given algae and answer the following questions:-



- (a) Name of algae ?
 (b) Type of sexual reproduction ?
 (c) Name of division ?
 (d) Stored food ?
 (1) *Laminaria* : Isogamous : Chlorophyta : Starch
 (2) *Volvox* : Oogamous : Chlorophyta : Starch
 (3) *Fucus* : Isogamous : Rhodophyta : Floridian starch
 (4) *Chara* : Oogamous : Rhodophyta : Glycogen
122. Select erroneous pair ?
 (1) Maize stem – Vascular bundles are scattered in parenchymatous ground tissue
 (2) Sunflower stem – Collenchymatous hypodermis as a homogenous layer
 (3) Maize root – Triarch xylem
 (4) Sunflower root – Tetrarch xylem
123. Which of the following is not a function of protein?
 (1) Glucose transport into cells
 (2) Fights infectious agents
 (3) Synthesis of sex-hormones
 (4) Synthesis of most of enzymes
124. Seedless tomatoes can be obtained by the use of-
 (1) Absciscic acid
 (2) Auxins
 (3) Cytokinins
 (4) Ethylene
125. Lymph differ from blood in possessing :-
 (1) only WBC
 (2) more RBC and WBC
 (3) more RBC and few WBC
 (4) more WBC and few RBC

126. Helicotrema is
 (1) An aquatic mammal
 (2) An aperture in between the III and IV ventricle of brain
 (3) An aperture in between two scala of cochlea of mammal
 (4) A disease of internal ear
127. In HGP, all the genes that are expressed as RNA referred to as-
 (1) Polypeptide sequence
 (2) Sequence annotation
 (3) Expressed sequence tag
 (4) Automated DNA sequencer
128. Most abundant sugar in honey is
 (1) Glucose (2) Fructose
 (3) Lactose (4) Sucrose
129. Darwin finches are not the example of
 (1) Genetic drift (2) Natural selection
 (3) Allopatric speciation (4) Adaptive convergence
130. Vas deferens arises from
 (1) cauda epididymis (2) caput epididymis
 (3) corpus epididymis (4) rete testis
131. Read the following statements regarding bryophytes and do as ask next to them
 (A) All cells of sporophyte undergo reduction division to produce haploid spores.
 (B) They play an important role in plant succession on barren rocks.
 (C) Plant body is thallus like and prostrate or erect and attached to the substratum by unicellular or multicellular rhizoids.
 (D) They prevent soil erosion.
 (E) Sex organs are multicellular. Male and female sex organs are called archegonium and antheridium respectively.
- How many of the above statements is/are not correct?
 (1) 2 (2) 3 (3) 1 (4) 4
132. Double stranded circular naked DNA is commonly observed in :-
 (1) only bacteria
 (2) only bacteria and chloroplast
 (3) only mitochondria and chloroplast
 (4) bacteria, mitochondria and chloroplast

133. How many carbon atoms are present in arachidonic acid excluding carboxyl carbon.

- (1) 18 (2) 19 (3) 20 (4) 21

134. If four enzymes, A, B, C & D with K_m values 20, 12, 32, 27 respectively, showing similar type of reactions then which of the following may be considered as most appropriate conclusion regarding these enzymatic reactions?

(1) In the reaction of enzyme 'C' minimum substrate

concentration is required to attain $\frac{V_{max}}{2}$

(2) Enzyme 'C' may be considered catalytically most active among all these enzymes

(3) Enzyme 'B' may be considered catalytically more active than enzyme A

(4) In the reaction of enzyme 'D' maximum substrate

concentration is required to attain $\frac{V_{max}}{2}$

135. Match the columns and choose the correct option:

Column-I		Column-II	
A	Salivary amylase	I	Proteins
B	Bile salts	II	Milk proteins
C	Renin	III	Starch
D	Pepsin	IV	Lipids

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-III, C-II, D-I

(4) A-I, B-II, C-III, D-IV

136. Consider the following statements

(A) The greenhouse effect is a natural phenomenon

(B) Tropospheric ozone is also known as "bad ozone"

(C) Eutrophication is the natural aging of a lake by nutrient enrichment of its water

(D) More than 2.5 micrometers particulate size cause the greatest harm to human health

How many of the above are incorrect?

- (1) 1 (2) 2 (3) 3 (4) 4

137. In bacteria, sigma factor is essential for

(1) Initiation of transcription

(2) Termination of transcription

(3) Synthesis of DNA

(4) Translation

138. Match the columns about diseases and causative agents:

	Column-I		Column-II
(a)	pneumonia	p.	Rhino viruses
(b)	common cold	q.	<i>Haemophilus influenzae</i>
(c)	filariasis	r.	<i>Microsporium</i>
(d)	ring worms	s.	<i>Wuchereria bancrofti</i>

(1) a-p, b-q, c-r, d-s (2) a-r, b-p, c-s, d-q

(3) a-q, b-p, c-s, d-r (4) a-p, b-q, c-s, d-r

139. Glandular tissue of each breast is divided into-

(1) 15-20 alveoli

(2) 15-20 mammary lobes

(3) 250-300 mammary lobes

(4) 1-3 alveoli

140. "Spermiogenesis" is a process in which :-

(1) spermatids change into spermatozoa

(2) spermatogonia produce a spermatid

(3) spermatocytes give rise to spermatozoa

(4) dormant spermatozoa become active just before ejaculation.

141. In brown algae sexual reproduction may be :-

(1) Isogamous

(2) Anisogamous

(3) Oogamous

(4) All

142. Read the following statement regarding eukaryotic cell and identify the true statements and select the correct option given below.

A. Duration of cell cycle varies from cell to cell and organism to organism.

B. Nucleus regulate the activity of cytoplasm by sending DNA in cytoplasm.

C. Cytoskeleton provide and maintain shape of cell

D. Endoplasmic reticulum divide the inter cellular space in two compartment

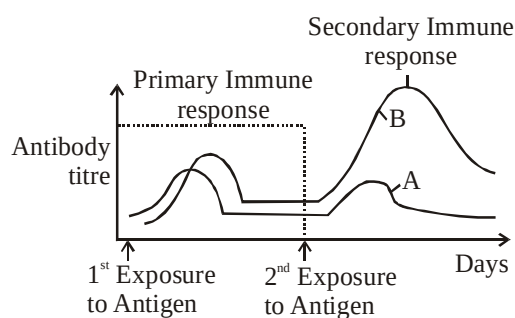
(1) only B and C (2) only B, C and D

(3) only A and C (4) A, B, C and D

143. Which of the following is a monomer unit of protein.

(1) Cystein (2) Cytosine (3) Thiamine (4) Uridine

144. Which one is correct for compound epithelium?
 (1) Multilayered – found in linings of duct, tubes & cavities
 (2) Single layered – mainly found in skin
 (3) Single layered – mainly helps in protection
 (4) Multilayered – mainly helps in protection
145. Diuresis is the condition in which :-
 (1) Kidney fail to excrete urine
 (2) Failure of urine production
 (3) Excretion of urine volume increase
 (4) Excretion of urine volume decrease
146. The most important cause of biodiversity losses is:-
 (1) Habitat loss and fragmentation
 (2) Alien species invasions
 (3) Co-extinctions
 (4) Over-exploitation
147. *Monascus purpureus* is source of-
 (1) Streptokinase (2) Statins
 (3) Cyclosporin-A (4) Lactic acid
148. (a) nasal congestion and discharge
 (b) sore throat, hoarseness, cough
 (c) headache, tiredness etc.
 are characters of -
 (1) malaria (2) typhoid
 (3) common cold (4) pneumonia
149. Regions outside the seminiferous tubules called interstitial spaces contain-
 (1) Leydig cells
 (2) Small blood vessels
 (3) Immunologically competent cells
 (4) All of these
150. Atretic follicle is
 (1) Which are not developed completely and degenerate
 (2) Other name of corpus luteum
 (3) Which extruded its oocytes
 (4) Graafian follicle
151. Members of monera are found normally
 (1) In such a habitat where temperature is very low
 (2) Where temperature is very high
 (3) Only in the body of living organism
 (4) These are cosmopoliton
152. Centromere division during mitosis occurs in :-
 (1) S phase (2) Telophase
 (3) Anaphase (4) Metaphase
153. During transport of food in plants, the process of loading at the source produces a.....A.....condition in the phloem and water from adjacentB.... moves into the....C....by.....D.....
 Choose the correct terms for A, B, C & D from the following options:
 (1) A = Hypotonic, B = Phloem, C = Xylem, D = Osmosis
 (2) A = Hypertonic, B = Phloem, C = Xylem, D = Active transport
 (3) A = Hypertonic, B = Xylem, C = Phloem, D = Osmosis
 (4) A = Hypotonic, B = Xylem, C = Phloem, D = Active transport
154. Dense regular connective tissue is:
 (1) Skin, proximal convoluted tubules
 (2) Skin, Ligament
 (3) Ligament, tendon
 (4) (2) & (3) both
155. Involuntary control of voluntary muscles is under the control of which part :-
 (1) Cerebrum (2) Medulla oblongata
 (3) Cerebellum (4) Midbrain
156. Among plants, which the following group is most species rich?
 (1) Angiosperms (2) Gymnosperms
 (3) Algae (4) Mosses
157. Which of the following characteristic is related to resistance to maize stem borer
 (1) Hairy leaves
 (2) Solid stem
 (3) Nector less variety
 (4) High aspartic acid
- Low nitrogen and low sugar content
- 158.



Labelled B is :-

- (1) IgM (2) IgG (3) IgD (4) IgA

- 159.** Here are some statements about oogenesis in females select the incorrect one-
- (1) First and second polar bodies are haploid cells
 - (2) Secondary oocyte and ovum are haploid cells
 - (3) Secondary oocyte and ovum are formed by unequal cell division
 - (4) During formation of primary oocyte from oogonia only mitosis occur but differentiation does not occur
- 160.** Which one of the following is the correct matching of the events occurring during menstrual cycle ?
- (1) Menstruation : Breakdown of myometrium and ovum not fertilised
 - (2) Ovulation : LH and FSH attain peak level and sharp fall in the secretion of progesterone
 - (3) Proliferative : Rapid regeneration of phase myometrium and maturation of Graafian follicle.
 - (4) Development of : Secretory phase and corpus luteum increased secretion of progesterone.
- 161.** Floridean starch is structurally related to :-
- (1) Chitin (2) Mannitol
 - (3) Laminarin (4) Amylopectin
- 162.** How many chromosomes are present in a diploid somatic cell of an organism during metaphase, if meiocyte in this organism has 23 bivalents during metaphase I ?
- (1) 23 (2) 46
 - (3) 92 (4) 69
- 163.** Deficiency of which of the following essential elements delay flowering in plants?
- (1) N, S, Mo (2) Ca, Mg, K
 - (3) Cu, Fe, Ni (4) Zn, Mn, Cl
- 164.** RBC formation in human body is not stimulated by:-
- (1) Thyroxine (2) Cortisol
 - (3) Erythropoietin (4) ANF
- 165.** Consider the following statements (a-d) and select the option which includes all the correct ones only:-
- (a) At the time of ventricular systole AV valves are closed and semilunar valves are opened.
 - (b) At the time of ventricular diastole AV valves are opened and semilunar valves are closed.
 - (c) At the time of joint diastole AV valves are opened and semilunar valves are closed.
 - (d) At the time of atrial systole AV valves are closed and semilunar valves are opened.
- Options :-**
- (1) Statements a, b and c
 - (2) Statements a and b only
 - (3) Statements c and d only
 - (4) Statements b, c and d
- 166.** Reasons for greater tropical biodiversity are :
- (1) Tropic remained undisturbed for millions of years
 - (2) More constant and predictable environment
 - (3) More solar energy
 - (4) All the above
- 167.** DNA precipitation out of a mixture of biomolecules can be achieved by treatment with-
- (1) Isopropanol
 - (2) Chilled chloroform
 - (3) Chilled ethanol
 - (4) Methanol

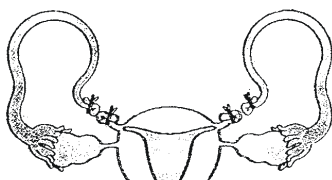
168. Which one the following group is **not** of infectious diseases:-

- (1) Cancer, Allergy, Rheumatoid arthritis
- (2) Dysentery, Plague, Diphtheria
- (3) Polio, Pneumonia, Tetanus
- (4) Smallpox, Ringworm, common cold

169. Which of the following ovarian hormone show two peaks during menstrual cycle ?

- (1) Estrogen
- (2) Progesterone
- (3) FSH
- (4) LH

170. What is the figure given below showing in particular?



- (1) Tubectomy
- (2) Vasectomy
- (3) Ovarian cancer
- (4) Uterine cancer

171. Select incorrect statement ?

- (1) Phellogen, inter-fascicular cambium and intra-fascicular cambium are example of lateral meristem
- (2) In the dicot root, vascular cambium is completely secondary in origin
- (3) Radial vascular bundles are found in roots.
- (4) In the dicot root, cork cambium is derived from epidermis

172. Match the column I with column II & select the correct option from option given below :-

	Column-I		Column-II
A	Quantasome	p	Mitochondria
B	Oxysome	q	Chloroplast
C	Cisternae	r	Golgi body
D	Microbodies	s	Ribosome
		t	Peroxisome

Options :

- (1) A-q, B-p, C-r, D-s
- (2) A-q, B-p, C-r, D-t
- (3) A-p, B-q, C-r, D-s
- (4) A-p, B-q, C-r, D-t

173. Root nodules of legume plants have which of the following adaptations which ensures that the nitrogenase enzyme is protected from oxygen?

- (1) Presence of α -ketoglutaric acid
- (2) Presence of ammonium ion
- (3) Presence of glutamic acid
- (4) Presence of leg-haemoglobin

174. Adrenal cortex hormone deficiency is:

- (1) Cretinism
- (2) Cushing disease
- (3) Dwarfism
- (4) Addison's disease

175. Presence of CO_2 in blood will decrease the pH because CO_2 combines with:-

- (1) H_2O to form H^+ and HCO_3^-
- (2) H_2O to form only HCO_3^-
- (3) H^+ ions to form HCO_3^-
- (4) H_2O to form only H^+ ions

176. Functional aspect of ecosystem is:-

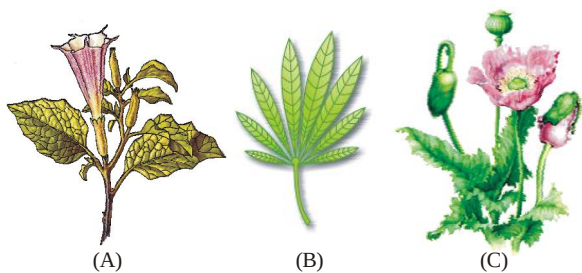
- (1) Stratification
- (2) Dominance
- (3) Decomposition
- (4) Species richness

177. Match the column I with column II

	Column-I		Column-II
(a)	i-gene	(i)	β -galactosidase
(b)	z-gene	(ii)	permease
(c)	a-gene	(iii)	repressor
(d)	y-gene	(iv)	transacetylase

- (1) a-iii, b-i, c-iv, d-ii
- (2) a-i, b-ii, c-iii, d-iv
- (3) a-ii, b-iii, c-iv, d-i
- (4) a-iv, b-iii, c-ii, d-i

178. Match these columns :-



	A	B	C
(1)	Opium	<i>Cannabis sativa</i>	Datura
(2)	<i>Cannabis sativa</i>	Datura	Opium
(3)	Datura	<i>Cannabis sativa</i>	Opium
(4)	Opium	Datura	<i>Cannabis sativa</i>

179. Placenta is connected to embryo through-

- (1) Endometrium
- (2) Umbilical cord
- (3) Yolk sac
- (4) Cervix

180. Select the hormone-releasing Intra-Uterine Devices :-

- (1) Vaults, LNG-20
- (2) Multiload 375, Progestasert
- (3) Progestasert, LNG-20
- (4) Lippes Loop, Multiload 375



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